NONVANISHING OF THE RESIDUES OF THE HEAT KERNEL FOR A LOGARITHMIC LAPLACIAN ON THE CIRCLE

KNOWLEDGE SENSE

Abstract. Rosenzweig and Stanfill showed that, on the diagonal, the fundamental solution (heat kernel) $K_1(t,\varphi,\varphi)$ of the Cauchy problem for a logarithmic Laplacian on the unit circle continues meromorphically in t, with a simple pole at $t=\frac{1}{2}$ and possible further simple poles at $t=\frac{1}{2}-m$, $m\geq 1$. The residue at $\frac{1}{2}-m$ is an explicit nonzero multiple of a rational number P_m, defined by ordinary Bell polynomials evaluated at Bernoulli numbers, and they asked whether $P_m\neq 0$ for every m. We prove that $8^m m!\,P_m$ is a 2-adic unit; hence $P_m\neq 0$ and $v_2(P_m)=-3m-v_2(m!)$ for all $m\geq 1$. Consequently, the poles of $K_1(t,\varphi,\varphi)$ are exactly the points $\frac{1}{2}-m$, $m\geq 0$. Elezović proved that, for a positive integer q, the coefficient of x^{-1} in the asymptotic expansion of $\exp(q\psi(x+s))$ does not depend on the real shift s. Combined with this, our theorem shows that this coefficient is nonzero for every odd positive integer q and every real s.

1. Introduction

Let A_1 be the nonlocal operator on the unit circle given by

$$A_1h(e^{i\varphi})=-\frac{1}{2}\int_0^{2\pi}\frac{h(e^{i\tau})-h(e^{i\varphi})}{|\sin(\frac{\tau-\varphi}{2})|}\,d\tau,$$

a logarithmic Laplacian on the circle; up to a constant shift and a multiplicative factor it is the conformal logarithmic Laplacian on $\mathbb{S}^1$ [4, Section 1]. Its eigenfunctions are $e^{\pm in\varphi}$ with eigenvalues $2\lambda_n$, where $\lambda_n=2H_{2n}-H_n$ and H_n is the nth harmonic number, $n\in\mathbb{N}_0$. The fundamental solution of $\partial_t u=-A_1u$, that is, the heat kernel of A_1, is

$$K_1(t,\varphi,\theta)=\frac{1}{2\pi}+\frac{1}{2\pi}\sum_{n\in\mathbb{Z}\setminus\{0\}}e^{in(\varphi-\theta)}e^{-2\lambda_{|n|}t},$$

and on the diagonal $\varphi=\theta$ the series converges for $\operatorname{Re}t>\frac{1}{2}$. Answering a question of Maz'ya recorded in [4], Rosenzweig and Stanfill proved that

Date: 13 October 2026.

2020 *Mathematics Subject Classification.* Primary 11B68; Secondary 11B73, 33B15, 35K08.

Key words and phrases. Bernoulli numbers, ordinary Bell polynomials, 2-adic valuation, digamma function, logarithmic Laplacian, fundamental solution, meromorphic continuation.

$K_1(t,\varphi,\theta)$ is entire in t for $\varphi \neq \theta$, and that $K_1(t,\varphi,\varphi)$ extends to a meromorphic function of t with a simple pole at $t = \frac{1}{2}$ and possible further simple poles at $t = \frac{1}{2} - m$, $m \in \mathbb{N}$ [4, Theorem 1.2]. To describe the residues we fix notation.

Bernoulli numbers are defined by $u/(e^u - 1) = \sum_{k\geq 0} B_k u^k/k!$, so that $B_1 = -\frac{1}{2}$ [2, Section 24.2(i)]. Put

$$s_k = (1 - 2^{1-k})\frac{2B_k}{k} \qquad (k \geq 1), \qquad S_1 = (s_k)_{k\geq 1}. \tag{1.1}$$

For a sequence $S = (s_k)_{k\geq 1}$ and $j \geq 0$ let

$$p_{j,S}(t) = \sum_{l=0}^{j} \frac{(-t)^l}{l!}\, \widehat{B}_{j,l}(s_1, s_2, \dots), \tag{1.2}$$

where

$$\widehat{B}_{j,l}(s_1, s_2, \dots) = \sum \frac{l!}{\prod_i a_i!} \prod_i s_i^{a_i}$$

and the sum runs over sequences $(a_i)_{i\geq 1}$ of nonnegative integers with $\sum_i a_i = l$ and $\sum_i i a_i = j$. Thus $\widehat{B}_{j,l}$ is the ordinary partial Bell polynomial [3], and (1.2) is [4, Definition 1.1]. Equivalently, as formal power series in z,

$$\sum_{j\geq 0} p_{j,S}(t)\, z^j = \exp\Big(-t \sum_{k\geq 1} s_k z^k\Big), \tag{1.3}$$

which is the generating function of the ordinary Bell polynomials [4, (2.6)]. Each coefficient in (1.3) is a finite sum, and no convergence of the series $\sum_k s_k z^k$ is involved. We write

$$P_m = p_{2m,S_1}\big(\tfrac{1}{2} - m\big) \in \mathbb{Q} \qquad (m \in \mathbb{N}_0),$$

so that $P_0 = 1$. By [4, Theorem 1.2, (1.7)], the residue of $K_1(t,\varphi,\varphi)$ at $t = \frac{1}{2} - m$ is

$$\frac{e^{-2\log 2 - \gamma}}{2\pi}\, e^{(4\log 2 + 2\gamma)m}\, P_m \qquad (m \in \mathbb{N}_0), \tag{1.4}$$

where γ is the Euler–Mascheroni constant. Rosenzweig and Stanfill conjectured that every point $\frac{1}{2} - m$ is an actual pole, which amounts to the following question.

Open Problem ([4, Open Problem 1.3])**.** *Show that* $p_{2m,S_1}(\frac{1}{2} - m) \neq 0$ *for all* $m \in \mathbb{N}$.

For a nonzero rational number x we write $v_2(x)$ for the exponent of 2 in x, that is, the exponent of 2 in the reduced numerator minus that in the reduced denominator. Let $\mathbb{Z}_{(2)}$ be the ring of rational numbers with odd denominator; its units are the rationals x with $v_2(x) = 0$. Our main result answers the question above and determines the 2-adic size of P_m.

Theorem 1.1. *For every integer $m \geq 1$, the number $8^m m!\, P_m$ is a unit of $\mathbb{Z}_{(2)}$. In particular, $P_m \neq 0$ and*

$$v_2(P_m) = -3m - v_2(m!).$$

Combining Theorem 1.1 with [4, Theorem 1.2] gives the pole structure conjectured there.

Corollary 1.2. *For every $\varphi \in [0, 2\pi)$, the poles of the meromorphic continuation of $t \mapsto K_1(t, \varphi, \varphi)$ are exactly the points $t = \frac{1}{2} - m$, $m \in \mathbb{N}_0$. All of them are simple, and the residue at $\frac{1}{2} - m$ is the nonzero number* (1.4).

The coefficients P_m also occur in the asymptotic expansion

$$e^{q\psi(x+s)} \sim x^q \sum_{n \geq 0} G_n(q, s)\, x^{-n} \qquad (x \to \infty,\ s \in \mathbb{R})$$

studied by Elezović [1], where ψ is the digamma function; Elezović writes p and t for the variables denoted here by q and s. He proved that, for a positive integer q, the coefficient $G_{q+1}(q, s)$ of x^{-1} does not depend on s [1, Theorem 3.1], and that it vanishes when q is even [1, Theorem 3.2]. In the notation of Rosenzweig and Stanfill these two facts appear as [4, Proposition 2.4 and Corollary 2.5], and [4, Remark 2.7] reformulates Open Problem 1.3 accordingly; see Section 5. We show there that $P_m = G_{2m}(2m - 1, \frac{1}{2})$, which gives the complementary statement for odd q.

Corollary 1.3. *Let q be a positive integer. Then $G_{q+1}(q, s) = 0$ for all $s \in \mathbb{R}$ if q is even, and $G_{q+1}(q, s) = P_{(q+1)/2} \neq 0$ for all $s \in \mathbb{R}$ if q is odd. Thus the term x^{-1} occurs in the asymptotic expansion of $e^{q\psi(x+s)}$ if and only if q is odd.*

Method of proof. The definitions (1.2)–(1.3), the analytic continuation of K_1 and the residue formula (1.4) are due to Rosenzweig and Stanfill [4], and the statements about $G_{q+1}(q, s)$ quoted above are due to Elezović [1]; we do not reprove them. The proof of Theorem 1.1 expands P_m as a finite sum over the partitions of m and uses two classical facts: the von Staudt–Clausen theorem, only in the weak form $v_2(B_{2r}) = -1$, and the fact that for a partition of m with a_r parts of size r the number

$$\frac{m!}{\prod_r r^{a_r} a_r!}$$

is an integer, since it counts the permutations of $\{1, \ldots, m\}$ of that cycle type. With these facts, the term of the partition into singletons is the unique term of least 2-adic valuation, which gives both $P_m \neq 0$ and the value of $v_2(P_m)$.

Organization. Section 2 collects the two classical inputs. Section 3 proves Theorem 1.1. Section 4 deduces Corollary 1.2, and Section 5 treats the digamma expansion and Corollary 1.3.

2. Preliminaries

Throughout, $v_2(0) = +\infty$, so that $\mathbb{Z}_{(2)} = \{x \in \mathbb{Q} : v_2(x) \geq 0\}$ and $2\mathbb{Z}_{(2)} = \{x \in \mathbb{Q} : v_2(x) \geq 1\}$. Every $x \in \mathbb{Z}_{(2)}$ with $v_2(x) = 0$ is a unit of $\mathbb{Z}_{(2)}$ and satisfies $x \equiv 1 \pmod{2\mathbb{Z}_{(2)}}$, since $a/b - 1 = (a-b)/b$ for odd a, b.

Lemma 2.1. *The numbers s_k of (1.1) satisfy $s_k = 0$ for every odd k. For $r \geq 1$,*

$$s_{2r} = \frac{2^{2r-1} - 1}{2^{2r-1}} \cdot \frac{B_{2r}}{r}, \qquad v_2(s_{2r}) = -2r - v_2(r).$$

In particular $s_2 = \frac{1}{12}$.

Proof. For $k = 1$ the factor $1 - 2^{1-k}$ vanishes, and $B_k = 0$ for odd $k \geq 3$ [2, Section 24.2(i)]. The displayed formula for s_{2r} is (1.1) with $k = 2r$. By the von Staudt–Clausen theorem [2, 24.10.1], $B_{2r} + \sum_p p^{-1} \in \mathbb{Z}$, where p runs over the primes with $(p-1) \mid 2r$. The prime 2 always occurs, and the contribution of the odd primes lies in $\mathbb{Z}_{(2)}$; hence $B_{2r} \in -\frac{1}{2} + \mathbb{Z}_{(2)}$ and $v_2(B_{2r}) = -1$. Since $2^{2r-1} - 1$ is odd for $r \geq 1$, we get

$$v_2(s_{2r}) = -(2r-1) + v_2(B_{2r}) - v_2(r) = -2r - v_2(r).$$

Finally $s_2 = \frac{1}{2} \cdot \frac{1}{6}$. □

For $m \geq 1$ let

$$\mathcal{A}_m = \Big\{ a = (a_1, \ldots, a_m) \in \mathbb{Z}_{\geq 0}^m : \sum_{r=1}^{m} r a_r = m \Big\},$$

the set of partitions of m recorded by their multiplicities, and put

$$N(a) = \frac{m!}{\prod_{r=1}^{m} r^{a_r} a_r!} \qquad (a \in \mathcal{A}_m). \tag{2.1}$$

Lemma 2.2. *For every $a \in \mathcal{A}_m$, $N(a)$ is the number of permutations of $\{1, \ldots, m\}$ having exactly a_r cycles of length r for each r. In particular $N(a)$ is a positive integer.*

Proof. This is classical [2, 26.4.7]; we recall the count. Consider the $m!$ arrangements of the labels $1, \ldots, m$ in a row. Cut each arrangement, from left to right, into a_1 blocks of length 1, then a_2 blocks of length 2, and so on, and read each block as a cycle. This gives a permutation of the required type, and every such permutation arises. A given permutation of this type arises from exactly $\prod_r r^{a_r} a_r!$ arrangements, since each cycle of length r can be written starting at any of its r entries, and the a_r cycles of length r can be listed in any order. Hence the number of such permutations is $m! / \prod_r r^{a_r} a_r! = N(a)$. □

3. Proof of Theorem 1.1

By Lemma 2.1, for every t the exponent $-t\sum_k s_k z^k$ in (1.3) with $S = S_1$ contains only even powers of z. Hence $p_{j,S_1}(t)$ vanishes identically for odd j, as observed in [4, Lemma 2.2].

Fix $m \geq 1$ and put $\alpha = m - \frac{1}{2}$, so that $-(\frac{1}{2} - m) = \alpha$ and $v_2(\alpha) = v_2(2m-1) - 1 = -1$. With $x = z^2$ we obtain

$$\sum_{j\geq 0} p_{j,S_1}\big(\tfrac{1}{2} - m\big) z^j = \exp\Big(\sum_{r\geq 1} c_r x^r\Big), \qquad c_r = \alpha\, s_{2r}.$$

Since

$$\exp\Big(\sum_{r\geq 1} c_r x^r\Big) = \prod_{r\geq 1}\sum_{a\geq 0} \frac{c_r^a\, x^{ra}}{a!},$$

comparison of the coefficients of x^m gives the finite expansion

$$P_m = \sum_{a\in\mathcal{A}_m} T(a), \qquad T(a) = \prod_{r=1}^{m} \frac{c_r^{a_r}}{a_r!}. \tag{3.1}$$

Since $\sum_r r a_r = m$ for $a \in \mathcal{A}_m$, multiplication by $8^m m! = m! \prod_r 8^{r a_r}$ gives

$$8^m m!\, P_m = \sum_{a\in\mathcal{A}_m} N(a) \prod_{r=1}^{m} b_r^{a_r}, \qquad b_r = 8^r\, r\, c_r, \tag{3.2}$$

with $N(a)$ as in (2.1). By Lemma 2.1,

$$v_2(b_r) = 3r + v_2(r) + v_2(\alpha) + v_2(s_{2r}) = 3r + v_2(r) - 1 - 2r - v_2(r) = r - 1.$$

Thus $b_r \in \mathbb{Z}_{(2)}$ for all r, $b_r \in 2\mathbb{Z}_{(2)}$ for $r \geq 2$, and $b_1 = 8\alpha s_2 = \frac{2m-1}{3}$ is a unit of $\mathbb{Z}_{(2)}$.

Let $a_* = (m, 0, \ldots, 0)$ be the partition into singletons. Its term in (3.2) is $N(a_*) b_1^m = b_1^m$, which is a unit. If $a \in \mathcal{A}_m$ and $a \neq a_*$, then $a_r \geq 1$ for some $r \geq 2$, so $\prod_r b_r^{a_r} \in 2\mathbb{Z}_{(2)}$; as $N(a) \in \mathbb{Z}$ by Lemma 2.2, the corresponding term of (3.2) lies in $2\mathbb{Z}_{(2)}$. Therefore

$$8^m m!\, P_m \equiv b_1^m \equiv 1 \pmod{2\mathbb{Z}_{(2)}},$$

so $8^m m!\, P_m$ is a unit of $\mathbb{Z}_{(2)}$. In particular $P_m \neq 0$, and taking valuations gives the formula for $v_2(P_m)$. □

For $m = 1$ the set $\mathcal{A}_1$ consists of a_* alone and $P_1 = c_1 = \frac{1}{24}$.

Remark 3.1. In terms of (3.1), the proof says that

$$T(a_*) = \frac{c_1^m}{m!} = \frac{1}{m!}\Big(\frac{2m-1}{24}\Big)^m$$

is the unique term of least 2-adic valuation. Indeed, with $L(a) = \sum_r a_r$, the identity $T(a) = N(a)\prod_r b_r^{a_r}/(8^m m!)$ gives

$$v_2(T(a)) - v_2(T(a_*)) = m - L(a) + v_2(N(a)),$$

and $m-L(a)=\sum_r (r-1)a_r \geq 1$ for $a \neq a_*$. By Legendre's formula $v_2(m!) = m-\sigma(m)$, where $\sigma(m)$ is the number of ones in the binary expansion of m, so Theorem 1.1 can also be written as $v_2(P_m) = \sigma(m) - 4m$.

Remark 3.2. The argument uses the point $t = \frac{1}{2} - m$ only through $v_2(t) = -1$. Hence, for every $t \in \mathbb{Q}$ with $v_2(t) = -1$ and every $m \geq 1$, the number $8^m m!\, p_{2m,S_1}(t)$ is a unit of $\mathbb{Z}_{(2)}$; here $b_1 = -2t/3$.

Remark 3.3. Differentiating (1.3) gives the recurrence

$$j\, p_{j,S}(t) = -t \sum_{k=1}^{j} k\, s_k\, p_{j-k,S}(t) \qquad (j \geq 1),$$

which is convenient for exact computation. The first values are listed below.

m	1	2	3	4	5
P_m	$\frac{1}{24}$	$-\frac{9}{640}$	$\frac{7625}{580608}$	$-\frac{4311167}{199065600}$	$\frac{27527769}{504627200}$
$v_2(P_m)$	-3	-7	-10	-15	-18

We also computed P_m in exact rational arithmetic for $m \leq 50$ and found agreement with Theorem 1.1; these computations are not used in the proof.

4. The poles of K_1 on the diagonal

We recall the part of [4, Proposition 2.8] that is needed. On the diagonal the kernel does not depend on φ:

$$K_1(t,\varphi,\varphi) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n \geq 1} e^{-2\lambda_n t}, \tag{4.1}$$

and the series converges absolutely and locally uniformly for $\operatorname{Re} t > \frac{1}{2}$. The eigenvalues have the asymptotic expansion [4, (2.21)]

$$2\lambda_n \sim 2\log n + s_0 + \sum_{k \geq 1} s_k n^{-k} \qquad (n \to \infty), \qquad s_0 = 4\log 2 + 2\gamma,$$

with s_k as in (1.1). By [4, Lemma 2.1, (2.1)], for every $M \in \mathbb{N}$ and every compact set $\mathcal{K} \subset \mathbb{C}$,

$$e^{-2\lambda_n t} = n^{-2t} e^{-s_0 t} \sum_{j=0}^{M-1} p_{j,S_1}(t)\, n^{-j} + e^{-s_0 t}\, O\big(n^{-2\operatorname{Re} t - M}\big)$$

uniformly for $t \in \mathcal{K}$. Summing over n for $\operatorname{Re} t > \frac{1}{2}$ yields [4, (2.19)]

$$K_1(t,\varphi,\varphi) = \frac{1}{2\pi} + \frac{e^{-s_0 t}}{\pi} \sum_{j=0}^{M-1} p_{j,S_1}(t)\, \zeta(j+2t) + F_M(t), \tag{4.2}$$

where F_M is π^{-1} times the sum over $n \geq 1$ of the remainder terms. Each remainder term is holomorphic in t. Let $\mathcal{K}$ be a compact subset of the half-plane $\operatorname{Re} t > \frac{1-M}{2}$ and put $\sigma_{\mathcal{K}} = \min_{t \in \mathcal{K}} \operatorname{Re} t$. By the bound above, the nth

remainder is bounded on $\mathcal{K}$ by $C_{\mathcal{K},M}\, n^{-2\sigma_{\mathcal{K}}-M}$, where the constant $C_{\mathcal{K},M}$ depends only on $\mathcal{K}$ and M. Since $2\sigma_{\mathcal{K}} + M > 1$, the series of remainders converges uniformly on $\mathcal{K}$. Hence F_M is holomorphic in the half-plane $\operatorname{Re} t > \frac{1-M}{2}$, and (4.2) gives the meromorphic continuation of $K_1(t,\varphi,\varphi)$ to this half-plane.[1]

Proof of Corollary 1.2. Fix $m \in \mathbb{N}_0$ and choose $M > 2m$, so that $t_m = \frac{1}{2} - m$ lies in the half-plane where F_M is holomorphic. In (4.2) the summand with index j is holomorphic except for a possible simple pole at $t = \frac{1-j}{2}$, coming from the pole of ζ at 1; as a function of t, $\zeta(j+2t)$ has residue $\frac{1}{2}$ there. The point t_m arises only for $j = 2m$, so

$$\operatorname*{Res}_{t=t_m} K_1(t,\varphi,\varphi) = \frac{e^{-s_0 t_m}}{2\pi}\, p_{2m,S_1}(t_m) = \frac{e^{-2\log 2-\gamma}}{2\pi}\, e^{(4\log 2+2\gamma)m}\, P_m,$$

which is (1.4). This number is nonzero: for $m = 0$ because $P_0 = 1$, and for $m \geq 1$ by Theorem 1.1. For odd j the polynomial p_{j,S_1} vanishes identically (Section 3), so the points $\frac{1-j}{2} \in \{0, -1, -2, \dots\}$ are regular. Since M is arbitrary, $K_1(t,\varphi,\varphi)$ is meromorphic on $\mathbb{C}$ and its poles are exactly the points $\frac{1}{2} - m$, $m \in \mathbb{N}_0$, all simple. □

Remark 4.1. Rosenzweig and Stanfill also study a second operator A_2 on $[-1,1]$, with a sequence S_2 given by $s_k^{(2)} = -B_k/k$ [4, (2.8)], where $B_1 = -\frac{1}{2}$. By [4, Proposition 2.4, (2.17) and Remark 2.7], [4, Open Problem 1.3] is equivalent to the nonvanishing of $p_{j+2,S_2}(-j)$ for all odd j, while [4, Open Problem 1.6(i)] is the same statement for even j. In fact $p_{2m+1,S_2}(1-2m) = -\frac{1}{2}P_m$; we derive this in (5.4) below, where we also comment on the convention for S_2 in [4, (2.11) and (2.17)]. The sign plays no role in these nonvanishing statements. Theorem 1.1 therefore settles the case of odd j. We do not address the case of even j, which concerns the operator A_2.

5. Exponentials of the digamma function

We follow Elezović [1], writing q and s for his variables p and t. For $s \in \mathbb{R}$, by [1, Theorem 2.3], the coefficients in $e^{q\psi(x+s)} \sim x^q \sum_n G_n(q,s) x^{-n}$ satisfy $G_0 = 1$ and

$$G_n(q,s) = \frac{q}{n} \sum_{k=1}^{n} (-1)^{k+1} B_k(s)\, G_{n-k}(q,s) \qquad (n \geq 1),$$

where $B_k(s)$ is the kth Bernoulli polynomial. Put $F(y) = \sum_{n\geq 0} G_n(q,s)\, y^n$. Multiplying the recurrence by ny^n and summing over n shows that it is the

[1] The proof of [4, Proposition 2.8] states the domination on $\operatorname{Re} t \geq \sigma$ with $Cn^{-2\sigma-M+1}$; the bound of [4, (2.1)] gives the exponent $-2\sigma_{\mathcal{K}} - M$ used here. Since M is arbitrary, either version suffices for the meromorphic continuation.

coefficientwise form of

$$yF'(y) = q\Big(\sum_{k\ge1}(-1)^{k+1}B_k(s)\,y^k\Big)F(y),$$

that is, the logarithmic derivative of F is $q\sum_{k\ge1}(-1)^{k+1}B_k(s)y^{k-1}$. Since $F(0)=1$, the recurrence is equivalent to the identity of formal power series

$$\sum_{n\ge0}G_n(q,s)\,y^n = \exp\Big(q\sum_{k\ge1}\frac{(-1)^{k+1}B_k(s)}{k}\,y^k\Big). \tag{5.1}$$

For $u\in\mathbb{R}$ let $S_2(u)=(-B_k(u)/k)_{k\ge1}$, as in [4, (2.11)]; thus $S_2(0)$ is the sequence S_2 of Remark 4.1. Since $B_k(1-s)=(-1)^kB_k(s)$, comparison of (5.1) with (1.3) gives

$$G_n(q,s) = p_{n,S_2(1-s)}(-q) \qquad (n\ge0,\ s\in\mathbb{R}). \tag{5.2}$$

Since $B_1(u)=u-\frac12$ and $B_k(1)=(-1)^kB_k$, the sequences $S_2(0)$ and $S_2(1)$ differ only in their first terms, $\frac12$ and $-\frac12$, so that $p_{n,S_2(1)}(t)=(-1)^np_{n,S_2(0)}(t)$ [4, Proposition 2.3]. Consequently, replacing $S_2(1-s)$ by $S_2(s)$ in (5.2) would change the signs of the terms of odd degree. In [4], the sentence following (2.11) identifies the sequence of (2.8) with $S_2(1)$; with the family (2.11) and $B_1=-\frac12$, the sequence of (2.8) is $S_2(0)$. Identities of odd degree must therefore be read with the appropriate member of the family; see (5.4) below. For a positive integer q and $j=q+1$, (5.2) reads

$$G_{q+1}(q,s) = p_{j,S_2(1-s)}(1-j).$$

Thus the independence of s in [1, Theorem 3.1] corresponds to [4, Proposition 2.4], the vanishing for even q in [1, Theorem 3.2] corresponds to [4, Corollary 2.5], and [4, Remark 2.7] states [4, Open Problem 1.3] as the nonvanishing of $p_{j,S_2(s)}(1-j)$ for even $j\ge2$.

At $s=\frac12$ the coefficients can also be expressed through S_1. By [4, (1.8)], $s_k=-\frac2kB_k(\frac12)$ for $k\ge1$. In particular $B_k(\frac12)=0$ for odd k, since $s_k=0$ for odd k by Lemma 2.1. Hence $(-1)^{k+1}B_k(\frac12)/k=s_k/2$ for every $k\ge1$, and comparison of (5.1) with (1.3) gives

$$G_n\big(q,\tfrac12\big) = p_{n,S_1}\big(-\tfrac q2\big) \qquad (n\ge0). \tag{5.3}$$

Taking $q=2m-1$ and $n=2m$ gives $P_m=G_{2m}(2m-1,\frac12)$.

The coefficient of the next degree can be computed in the same way. For a positive integer q, the translation formula [1, (3.1)] with $n=q+2$ reads $G_{q+2}(q,s+\tau)=G_{q+2}(q,s)-\tau\,G_{q+1}(q,s)$, because the binomial coefficients $\binom{k-2}{k}$ vanish for $k\ge2$. Take $q=2m-1$ and $s=\frac12$. Then $G_{2m+1}(2m-1,\frac12)=0$ by (5.3), since p_{j,S_1} vanishes identically for odd j, and $G_{2m}(2m-1,\frac12)=P_m$. With (5.2) and $u=1-s$, this gives

$$p_{2m+1,S_2(u)}(1-2m) = \big(u-\tfrac12\big)P_m \qquad (m\ge1,\ u\in\mathbb{R}). \tag{5.4}$$

For $u = 0$, that is, for the sequence of [4, (2.8)], this is the identity $p_{2m+1,S_2}(1-2m) = -\frac{1}{2}P_m$ of Remark 4.1. The relation [4, (2.17)] with the factor $-\frac{1}{2}$ is printed for $S_2(1)$; by (5.4) it holds for $S_2(0)$, while for $S_2(1)$ the sign is opposite. For example, $p_{3,S_2(0)}(-1) = -\frac{1}{48}$ and $p_{3,S_2(1)}(-1) = \frac{1}{48}$, with $P_1 = \frac{1}{24}$.

Proof of Corollary 1.3. For a positive integer q, the coefficient $G_{q+1}(q,s)$ does not depend on $s \in \mathbb{R}$ [1, Theorem 3.1]; see also [4, Proposition 2.4] and (5.2). If q is even, it vanishes [1, Theorem 3.2]; see also [4, Corollary 2.5]. If $q = 2m-1$ is odd, then by (5.3) it equals $G_{2m}(2m-1, \frac{1}{2}) = P_m$, which is nonzero by Theorem 1.1. Since $x^q \cdot x^{-(q+1)} = x^{-1}$, the last statement follows. □

Remark 5.1. For $q = 3$, the table in [1, p. 4] lists $G_4 = -\frac{9}{640}$, in agreement with P_2 and with the independence of s. Inserting $(-1)^{k+1}B_k(\frac{1}{2}) = ks_k/2$ into [1, Theorem 2.3] and using Lemma 2.1, one obtains the recurrence

$$G_{2n}\big(q, \tfrac{1}{2}\big) = \frac{q}{2n}\sum_{k=1}^{n}\big(1 - 2^{1-2k}\big)B_{2k}\, G_{2n-2k}\big(q, \tfrac{1}{2}\big),$$

which gives $G_2(q, \frac{1}{2}) = \frac{q}{24}$. The recurrence printed as [1, (4.1)] differs from this one in sign and in the power of 2; the argument above does not use it.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] N. Elezović, *Asymptotic expansions of exponentials of digamma function and identity for Bernoulli polynomials*, preprint (2013), arXiv:1312.1604v1.

[2] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.8 of 2026-09-15, `https://dlmf.nist.gov/`.

[3] C. O'Sullivan, *De Moivre and Bell polynomials*, Expo. Math. **40** (2022), no. 4, 870–893; arXiv:2203.02868.

[4] B. Rosenzweig and J. Stanfill, *On the fundamental solutions of two nonlocal parabolic equations related to logarithmic Laplacians*, preprint (2026), arXiv:2606.04225v1.

Knowledge Sense
Email address: `math@knowledgesense.jp`